Signal Intelligence Dashboard

A build brief for Ben

Date	29 March 2026
Asset	Gold (XAU/USD) — starting point
Status	Ready to start

What this is

I want to build a signal monitoring dashboard. The idea is a simple live, private web dashboard that shows buy and sell signals for gold, tells us what regime the market is in (uptrend or downtrend), stacks multiple indicators together so we can see confluence, and fires an alert to our phones or email when something meaningful happens.

Nothing like this exists for us yet. Broker platforms offer some overlap most have charts, some have basic alerts but their primary model is charting, and that complexity gets in the way. What we are building is different: the whole point is simplicity. Clean signal cards, stacked indicators, and a clear read on market regime at a glance. No noise, no digging through charts to figure out what the market is doing. I want to get it done now because with the tools we have it is genuinely quick to build, and having it running before we need it badly is far smarter than scrambling to put something together when we are already live on a funded account.

The bigger picture

I can see us running a funded account in the range of £200k. At that size, getting signals right — knowing when to get in, when to stay out, and when a regime has shifted — is the difference between trading well and trading blind. This is the tool that makes us better.

Gold is where we start. It is liquid, macro-driven, and something we have a clear view on. Once we have it working for gold, the same platform becomes the home for every other signal or indicator idea the team comes up with as we grow but we don't need to think about that now. Let's just get gold right.

What it does

The dashboard receives signals from TradingView when they fire, stores them, and shows everything in a clean live interface. The trader looks at the dashboard, sees the current signal and regime, and makes the call. No automation in this version — just clean, timely intelligence.

The signals we want on it

- **Gold buy / sell signal:** the primary call — buy, sell, or hold, with a confidence score based on how many indicators agree
- **Regime indicator:** is gold in an uptrend or a downtrend right now, and when did that change — this is crucial for knowing whether to be buying dips or selling rallies
- **Signal stack:** show each indicator side by side so we can see when they are all pointing the same way — EMA alignment, ADX reading, RSI zone, momentum direction
- **Alert history:** a full log of every signal that fired with timestamp and price, so we can review performance over time

What fires an alert

- **New buy or sell signal:** pushes to Telegram and optionally email the moment it fires
- **Regime switch:** uptrend flipping to downtrend or vice versa — these are the important ones, we want to know immediately
- **High confluence:** when three or more indicators all agree on direction at the same time

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The stack

I want to keep this simple. The architecture below is the most proven real-world setup for exactly this kind of system and it is fast to build. No over-engineering — just the right tools for the job.

Layer	Tool	Why
Signals	**TradingView**	We define the signal logic here, verify it visually on the gold chart, and set up alerts that fire when conditions are met.
Transport	**Webhook**	When a TradingView alert fires it sends an HTTP POST to our server. Simple and reliable — no polling, no delays.
Backend	**Python + FastAPI**	Receives the webhook, processes the signal, applies regime logic, and saves everything to the database.
Database	**SQLite / Postgres**	Stores every signal with timestamp and price. Start with SQLite — it is zero setup and more than enough for this.
Dashboard	**Streamlit**	Fast to build, looks good, supports live refresh. The whole team can access it via a URL.

Alerts	**Telegram bot**	Signals pushed to a private team channel the moment they fire. No one needs to be watching the screen.
Hosting	**Cloud server**	Runs 24/7. Not on anyone's laptop — on a small always-on server so it never goes down.

In one line

TradingView alert fires → webhook hits our Python server → signal saved to database → dashboard updates → Telegram message sent.

What the dashboard looks like

Clean and functional. This is an internal tool, not a product, so we prioritise clarity over aesthetics. Here is what I want on it:

Live signal panel

- **Current regime:** big, colour-coded — green for uptrend, red for downtrend, grey for unclear
- **Active signal:** BUY / SELL / HOLD with confidence score and time of last signal
- **Signal stack:** individual indicator readings so we can see what is driving the call

Regime switch feed

- Shows when the last regime change happened and at what price
- Highlights if a switch has occurred recently so it is impossible to miss

Signal history table

- Every signal that has fired — asset, direction, type, time, and price
- Filterable and downloadable so we can track how signals are performing

System health

- Simple indicators showing the webhook, database and alert system are all running fine

How I want to build it

Step by step. Each one working and tested before we move to the next.

1. Define the gold signal logic in TradingView. EMA structure, ADX, RSI zones, regime classification. Verify it looks right on the chart.
2. Stand up the Python backend with FastAPI. Create the webhook endpoint and confirm it receives TradingView payloads correctly.

3. Wire up the TradingView alert to the webhook. End-to-end test: alert fires, payload arrives, gets parsed.
4. Set up the database and signal table. Asset, regime, direction, confidence, timestamp, price.
5. Build the Streamlit dashboard. Live signal panel and history log first. Refresh every 30 seconds.
6. Add the Telegram bot. Signal fires, message goes to team channel.
7. Add the regime switch indicator to the dashboard.
8. Run it live on gold for a week, review what is working, and decide where to go next together.

What we need at the end of this

Webhook endpoint	Python server on a cloud VPS receiving TradingView alerts
Signal database	Full signal history stored from day one
Live dashboard	Streamlit app, password protected, accessible to the team
Telegram alerts	Bot set up and tested, pushing signals to team channel
TradingView setup	Gold signal logic in Pine Script, alert connected to the webhook
Regime indicator	Live regime state on the dashboard with switch history

A few things I want to keep in mind

- **No auto-trading:** not in this version — we see the signal and we make the call
- **Keep it simple:** the first version should be running fast, not perfect — we improve it as we go
- **Always on:** cloud server, 24/7, not dependent on anyone's laptop
- **Built to grow:** every signal is a module so we can add new ones easily as the team evolves

That is it. Happy to jump on a call or go back and forth on any of this. I want to get started as soon as you are ready — let me know what you need from me to kick it off.

One more thing worth noting —
If we build this well, the platform itself has commercial potential down the line. A clean, reliable signal dashboard with good indicators is exactly the kind of thing retail traders and small funds would pay monthly access for. That is not the focus right now — right now the focus is building something that gives us an edge, makes us better informed, and helps us execute with more confidence. But it is worth keeping in the back of our minds as we design it. If we build it right from the start, that door stays open.

Internal use only.